Hydrocephalus Outcomes Among Spanish Speaking Patients at a Single Institution

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Background

- Hydrocephalus is a condition in which there is a buildup of cerebrospinal fluid in the brain and is the most common pediatric neurosurgical condition
- Prior studies have demonstrated worse outcomes in hydrocephalus among racial minorities and those with lower socioeconomic status; however, to our knowledge, no study has examined outcomes among patients speaking non-English languages
- In studies conducted in other pediatric subspecialties, up to 32% of Spanish-speaking caregivers felt their child would have received better care if they spoke English as a primary language
- Our study's aims were two-fold:
  1. Determine if there are differences in hydrocephalus outcomes (e.g., healthcare utilization, complications rate) among English-speaking patients (ESPs) and Spanish-speaking patients (SSPs)
  2. Determine if there are differences in postoperative communication patterns between parents/caregivers and the Pediatric Neurosurgery clinic among ESPs and SSPs

Methods

- The LPCH electronic medical record was queried to identify patients with a ventriculoperitoneal shunt (VPS) or past endoscopic third ventriculostomy (ETV) from 2012-2022
- Fifty ESPs and fifty SSPs aged < 18 years old were randomly selected
- Chart review was performed to collect data
- The following variables were collected:
  - Demographics
  - Healthcare utilization (e.g., length of stay and emergency department (ED) visits)
  - Postoperative complications
  - Communication patterns with clinic (e.g., number of calls/messages three months postoperatively)
- Multiple linear regressions were run to assess the significance of differences between SSPs and ESPs
- Analysis conducted using Stata/IC 16.1

Results

- SSPs were more likely to have a comorbidity than ESPs (p=0.05)
- SSPs were more likely to have a VPS (p=0.01) compared to an ETV
- Number of messages/calls was significantly associated with language, but not with comorbidity, insurance, or procedure

Healthcare utilization outcome	English-Speaking Patients	Spanish-Speaking Patients	P-values
Length of stay (average days)	7.3	21.5	**<0.01***
% of patients with ED visits within 30 days	12%	22%	0.29
% ED visits related to hydrocephalus	43%	46%	0.77
% of patients with readmission within 30 days	10%	34%	**0.01***
% of readmissions related to hydrocephalus	60%	35%	0.61
% of patients with a reoperation within 30 days	14%	14%	>0.99

Table 1. The healthcare utilization of ESPs versus SSPs. SSPs are statistically significantly more likely to have a longer length of stay and a readmission within 30 days of discharge.

Communication outcome	English-Speaking Patients	Spanish-Speaking Patients	P-values
% patients making at least one phone call or sending one message	58%	16%	**<0.01***
Total number of calls/ messages sent	58	10	**<0.01***
Outcome from call/ message			
• Reviewed instructions and supportive care	9	3	0.36
• Medical course change (e.g., making new/ next-day appointment, prescribing new medication, placing new referral)	17	0	**0.05***
• Reassurance	16	2	>0.99
• Sent to ED	3	4	**0.01***
• Logistics (e.g., rescheduled appointment)	13	1	0.67

Table 2. The differences in communications between ESP and SSP parents/caregivers, with communication outcomes. The most significant outcomes were medical course change and whether the patient was sent to the ED.

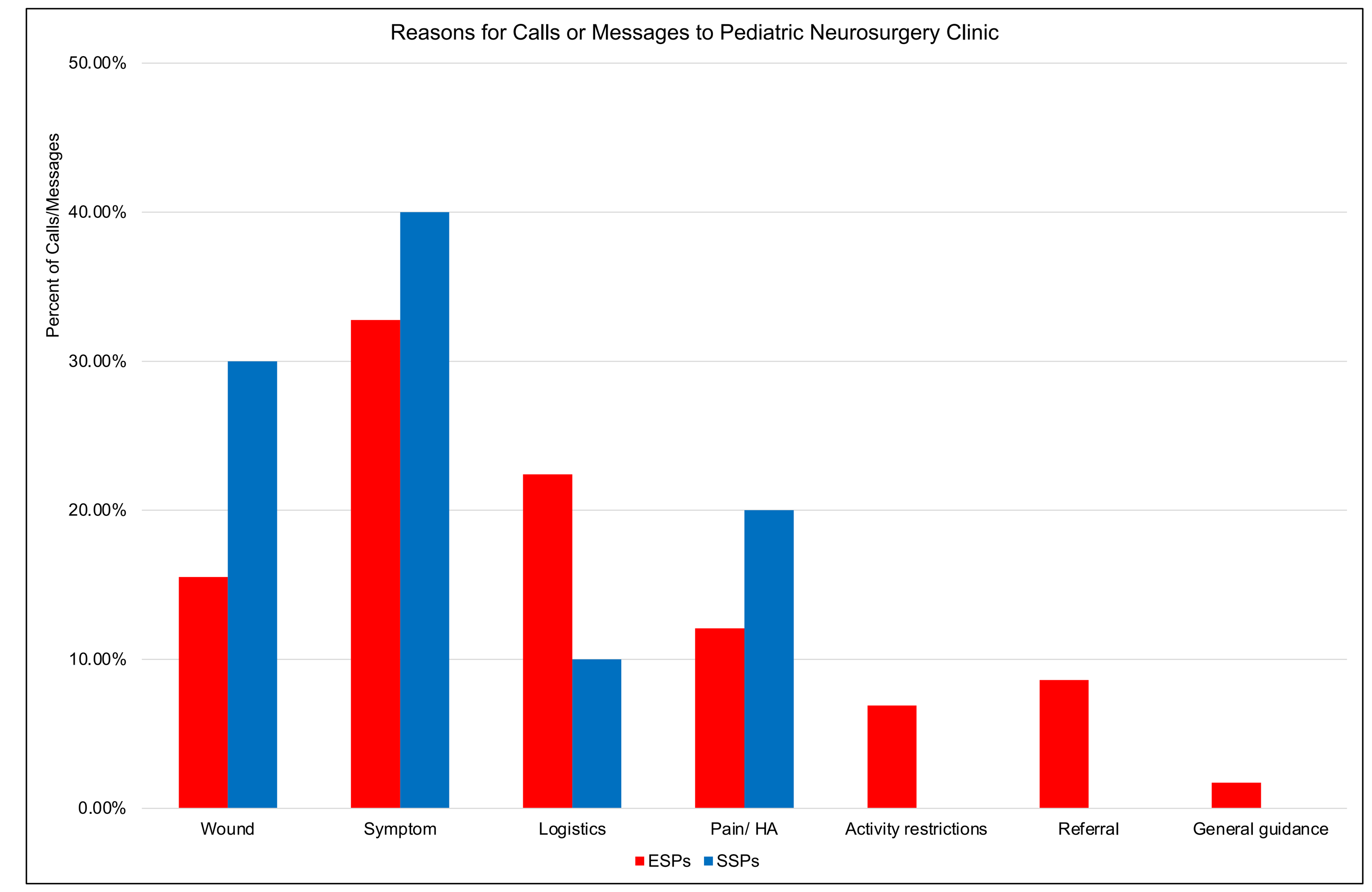

Figure 1. The reasons patient caregivers messaged or called the neurosurgery clinic within 3 months of surgery discharge. Despite the statistically significant difference in the % of SSPs who contact the hospital versus the % of ESPs who do, the reasons for their contact are not significantly different.

Conclusion

- SSPs were statistically significantly more likely to have a comorbidity, longer length of hospital stay, and all-cause readmission in children undergoing a procedure for hydrocephalus compared to ESPs.
- Only 16% of SSPs communicated with clinic postoperatively compared to 58% of ESPs. This is an important finding, as our findings suggest that SSPs are more medically complex patients but communicate with the medical team at significantly lower rates than their ESP counterparts.
- The difference in communication could be due to language barriers.
- Further research should explore communication barriers and preferences among SSPs to ensure equal access to neurosurgical care beyond the inpatient stay.

References

- Jin, Diana L., et al. "Cross-sectional analysis on racial and economic disparities affecting mortality in preterm infants with posthemorrhagic hydrocephalus." *World Neurosurgery* 88 (2016): 399-410.
- Kulkarni, Abhaya V., et al. "Medical, social, and economic factors associated with health-related quality of life in Canadian children with hydrocephalus." *The Journal of pediatrics* 153.5 (2008): 689-695.
- Pendleton, Courtney, et al. "Posthemorrhagic hydrocephalus in preterm neonates: socioeconomic characteristics in a single-institution experience." *Pediatric Neurosurgery* 48.2 (2012): 80-85.
- Stephen JM, Zoucha R. "Spanish Speaking, Limited English Proficient Parents whose Children are Hospitalized: An Integrative Review." *J Pediatr Nurs.* (2020): 52:30-40